

Unit 1 – Exploring One Variable Statistics
 HW 1 – Categorical Data

Name: _____

1) Below is a list of unorganized data from the 2020 election. This gives the state, their geographical location, and the voting majority.

<i>State</i>	<i>Location</i>	<i>Electoral College Vote in 2020</i>
AL	South	Republican
AK	West	Republican
AZ	West	Democrat
AR	South	Republican
CA	West	Democrat
CO	West	Democrat
CT	Northeast	Democrat
DE	South	Democrat
FL	South	Republican
GA	South	Democrat
HI	West	Democrat
ID	West	Republican
IL	Midwest	Democrat
IN	Midwest	Republican
IA	Midwest	Republican
KS	Midwest	Republican
KY	South	Republican
LA	South	Republican
ME	Northeast	Democrat*
MD	South	Democrat
MA	Northeast	Democrat
MI	Midwest	Democrat
MN	Midwest	Democrat
MS	South	Republican

MO	Midwest	Republican
MT	West	Republican
NE	Midwest	Republican*
NV	West	Democrat
NH	Northeast	Democrat
NJ	Northeast	Democrat
NM	West	Democrat
NY	Northeast	Democrat
NC	South	Republican
ND	Midwest	Republican
OH	Midwest	Republican
OK	South	Republican
OR	West	Democrat
PA	Northeast	Democrat
RI	Northeast	Democrat
SC	South	Republican
SD	Midwest	Republican
TN	South	Republican
TX	South	Republican
UT	West	Republican
VT	Northeast	Democrat
VA	South	Democrat
WA	West	Democrat
WV	South	Republican
WI	Midwest	Democrat
WY	West	Republican

*Split Votes

a) Construct a bar graph of the amount of states in each region of the US.

b) Construct a bar graph of the amount of states who voted either republican or democrat in the 2020 election.

c) Create a two way table showing the distribution of region and political party.

2) You measure the age, marital status and earned income of a random sample of 1463 women. The number and type of variables you have measured is

- (A) 1463; all quantitative.
- (B) four; two categorical and two quantitative.
- (C) four; one categorical and three quantitative.
- (D) three; two categorical and one quantitative.
- (E) three; one categorical and two quantitative.

3) As part of a survey of college students, a researcher is interested in the variable class standing. She records a 1 if the student is a freshman, a 2 if the student is a sophomore, a 3 if the student is a junior, and a 4 if the student is a senior. The variable class standing is

- (A) Categorical
- (B) Quantitative
- (C) Quantitatively Categorical
- (D) All of the above